DEPARTMENT OF BIOTECHNOLOGY

Faculty Name **:** Mr. R. Gnanasekaran

Faculty Code **:** HTS 1342

Subject Name **:** Bioprocess Principles

Subject Code **:** BT8404

Year & **Semester** **:** II & IV

VEL TECH HIGH TECH
Dr. RANGARAJAN Dr. SAKUNTHALA ENGINEERING COLLEGE
(Approved by AICTE, NEW DELHI. Affiliated to Anna University, Chennai)

DEPARTMENT OF BIOTECHNOLOGY

COURSE DETAILS

Faculty Name: Mr. R. Gnanasekaran **Faculty Code:** HTS1342

Subject Name: Bioprocess Principples **Subject Code:** BT8404

Department: Biotechnology **Year** & **Semester:** II & IV

COURSE OUTCOMES

At the end of the course, the student will be able to

CO No	Course Outcomes	Knowledge Level
C214.1	***Summarize***engineering principles to systems containing biological catalysts to meet the needs of the society.	K1
C214.2	***Understand*** the promises of molecular biology and genetic engineering into new processes to make bio-products in economically feasible way	K2
C214.3	***Discuss*** about the death kinetics of microorganism for both batch and continuous process.	K2
C214.4	***Elaborate*** the biological materials to improve its usefulness by finding the optimal formulation materials to facilitate product production.	K2
C214.5	***Interpret*** the kinetics of living cells and to develop a strategy to solve the issues emerging during fermentation processes	K3

Mapping of Course Outcomes with Program Outcomes and Program Specific Outcomes

BT6405	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3	PSO4
C214.1	3	-	-	-	-	1	1	-	-	-	-	2	3	3	-	-
C214.2	3	-	-	-	-	1	2	-	-	-	-	-	3	3	-	-
C214.3	3	3	1	-	-	-	2	1	-	-	-	-	-	3	-	-
C214.4	3	3	-	-	-	1	2	1	-	-	-	3	3	-	-	-
C214.5	3	3	1	-	-	1	2	1	-	-	-	3	3	-	-	-

BT6405	PO1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3	PSO4
C214	3	3	1	-	-	1	2	1	-	-	-	3	3	3	-	-

K1 – Remember; K2 – Understand; K3 – Apply; K4 – Analyse; K5 – Evaluate; K6 - Create

Mapping Relevancy

1: Slight (Low) 2: Moderate (Medium) 3 Substantial (High) - : No correlation

Unit I INTRODUCTION TO BIOPROCESS

PART – A QUESTIONS AND ANSWERS

1.What are the 3 different divisions in Biotech historical development?
Periods of Ignorance, Discovery, Industrial development

2.What are the areas of operations for bioprocess engineer in a small scale industry
Utilities, fermentations, downstream, culture maintenance and development, QC and packing

3.What are intracellular products? (April / May 2012)
Recombinant products produced inside the cell and retained throughout the process

4.What are the 4 stages of fermentation industry?
Utilities, fermentation bay, downstream process, packaging.

5.State 4 important chromatographic separations
Affinity, Ion exchange, Gel filtration, HPLC

6.State 2 importance of flow sheeting
To know the arrangement of equipments, process stream connections.

7.What are the Utilities used in fermentation industry
DM water, Soft water, Steam, Chill water, air

8.What is meant by pasteurization (Nov / Dec 2012)
Partial sterilization by increasing temperature up to its boiling point and cooling

9. What are the two major divisions in Biotechnology?
Traditional Biotechnology, New era biotechnology

10. Name a high value low volume products of Biotechnology
Therapeutic proteins, erythropoietin, monoclonal antibodies etc.

11. What are the areas of operations for bioprocess engineer in a medium scale industry
Fermentation, Utilities, downstream operations, microbiology labs.

12. What are the areas of operations for bioprocess engineer in a larger scale industry
Fermentation and downstream operations.

13. What is upstream processing?
Operations to prepare fermentation process, Steam generations, compressed air generations

14. What is downstream processing?
Operations to prepare the fermentation products, Centrifugation, Cell disintegrations

15. What are extracellular products? (April/May 2012)
The cellular products produced and excreted to the medium from cell

16. Define flow sheeting.
Diagrammatic model of the Bioprocess, Block diagram, Pictorial representation

17. What are the Standards for pictorial representation (Nov / Dec 2012)
British standard BS 1553, American national standard Institute (ASNI)

18. What are the Utilities used in fermentation industry
DM water, Soft water, Steam, Chill water, air

19. Explain the geometrical ratios of L/D, Impeller Dia/D
L/D = 0.72; Impeller Dia/D = 0.4

20. What is importance of aeration and agitation (April/May 2012)
To provide sufficient oxygen for metabolic requirement, and to ensure uniform suspension of microbial cells as homogeneous nutrient medium

PART B QUESTIONS

1. **Mention the detailed sketch of Outline of Integrated Bioprocess with all unit operations.**
   Draw the schematic diagrammatic flow chart of the overall process from Raw material procurement to product recovery and effluent treatment. The flow chart should represent Medium raw material – Medium formulations – Sterilization- Stock culture – Shake flask level – seed fermentor – production fermentor – culture broth – cell separation – Biomass – cell lysis – Product extraction – product purification – product packaging and effluent treatment etc.

2. **Describe briefly the concept of designing a fermenter. What factors do you consider as essential for a successful design and operation of a fermenter (Nov/Dec 2011)**

A clear schematic representation of the vessel with steam entry and exit points, chill water circulation, impeller and aerators and its position, top plate with septum ports, sensors and its position, sampling and drain ports and their usages.

3. **What are the various parameters that you need to monitor and control for the successful operation of a fermentor. (Nov/Dec 2011)**

   The Physical and chemical parameters to be monitored is to be listed first. The concept of Inline sensor, online sensor, Offline sensor to be explained in details with types of measurements. Selection criteria for the sensor for each parameters to be analyzed as response time, gain, sensitivity, accuracy, ease, speed of calibration, stability, precision

   Temperature: types like Mercury in glass, bimetallic, Pressure bulb, Metal resistance or thermisters to be explained with diagram and principle of operation

   Flow measurement and Control: Gases, rotameter types and principle of measurement, Thermal mass flow control with operational principle, calculation of flow with reference to temperature difference, electrical flow transducers, load cell etc, Metering pumps, types, operational principles

   Pressure measurement: Bourdon tube pressure gauge with neat diagram, Diaphragm type pressure sensor used, Agitator shaft power by watt meter and rate of stirring by tachometer, Foam measurement and control by simple circuit loop. The importance of controlling dissolved oxygen, and Galvanic and Polarographic type of probes and its material of construction with electrolyte assembly. The concept of pH probe used in dissolved carbondioxide measurement to be explained with a measure of one unit pH change when 10 folds of dCO_2 raise to be stated.

4. **Explain the configurations of the Bioreactor and its relative advantages (April / May 2010)**

   Other configurations of a fermentor for the specific purposes, Tower fermentor: Height; diameter is important with 6:1- bubble column air spurge at base even 16:1 with perforated plates positioned at intervals in the tower for maximum yeast production. Cylindro- conical vessels SS vertical tube with hemispherical top and a conical base with a included angle 70 dec. – agitated and design provides reduced process time and movement of fluid, sedimented yeast can be easily removed Air lift fermentor: air is the driving force- draw tube positioned, air enrich media, and low air media displacement- cultures grown with less shear force. 3 compartment multiple air lift loop fermentor- 3 draw tube assembly. The hydrodynamic fermentor types as deep jet model and loop model, cyclone column with neat diagram and principle of operations. Packed bed column and Rotating disc fermentor for immobilized cell system.

5. **What are the general requirements of a fermentation process? (Nov/Dec 2011)**

   Refer to the class notes and explanations related to Bioreactor and its parts. A clear schematic representation of the vessel with steam entry and exit points, chill water circulation, impeller and aerators and its position, top plate with septum ports, sensors and its position, sampling and drain ports and their usages. The service line equipments such as steam generation, chill water circulation, air compressor, deionized water, autoclaving, laminar flow chamber etc to be explain with uses.

6. **Explain about the different types and principles of Flow sheeting system. (Nov/Dec 2010)**

   It is the diagrammatic representation of the process into more explanatory ways. To know arrangement of equipment and Process stream and Process connection and process stream flow rate and opearation condidtion. Types as Block diagram in boxes, adnd Pictrorial representation as Piping and instrumentational diagram and Process flow diagaram and internation standards and norms for figurative notations to be explained with example.

7. **Write a detailed note on Material of construction and selection in fermentor fabrication.**

   Strict aseptic conditions- repeated steam sterilization-small scale possible to use glass stainless steel- 2 basic types used (i) A glass vessel with round flat bottom and top SS flanged carrying plate- borosilicate battery jars used (ii) A glass cylinder with SS top and bottom plates, sterilized insitue, max capacity is 30 cm. Material of construction should withstand pressure, Sterilization, Corrosion, potential toxicity, cost etc. AISI standard, Stainless steel composition, areas where wood, plastic and concrete used, corrosion resistance, thin hydrous oxide film on the surface of the metal, stabilized by chromium and non-porous, insoluble self healing material. Increased chromium content increases resistance, presence of molybdenum increases resistance of SS to halogen salts and seawater- electro polishing done, and Thickness of construction varies with size of the fermentor.

8. **Discuss the types of seal system in fermentation (Nov/Dec 2010)**

   Aseptic requirement of fermentation- joints of glass to glass, glass to metal and metal to metal, glass to metal joins with gasket, lip seal and O ring type diagram and explain each, Nitryl /butyl rubbers used for seal, double O ring with steam lock according to level of operations. Explain the principle and operations of stuffing box packed gland seal, simple bush seal, and mechanical seal and magnetic drives, discuss its advantages and disadvantages

9. **Explain the various Bioprocess development of Antibiotic era & Post antibiotic era**

Periods of Bioprocess development: Period of ignorance (pre 1800), Period of Discovery (1800 -1900) and Period of industrial development (post 1900) Points related to Traditional Biotechnology, New era biotechnology, Different products in Biotechnology, History division in to Period of Ignorance: Domestic development, Wine, Brewing etc, Period of Discovery, invention of microscope, Pasteur and fermentation,

Leibig findings, Period of Industrial development: Antibiotic development as Emmerich and low and Alexander fleming, Recent development qualitative and quantitative development etc

10. **With the help of case study, explain the general requirements of industrial scale fermentation processes. (Nov / Dec 2012)**

    Up stream processing: Utilities such as steam, Chill water, Air supply to be explained, media procurement, media formulation and medium sterilization to be explained Fermentation and condition to the production to be explained **Downstream processing;**The sequential steps in separation of active principle from fermentation broth upto product packing. In the down stream process the unit operation cell lysis techniques ether by ultrasonification, bead beating, or French press cell lysis etc i.e Broth and biomass separation using Centrifugation or filtration, Cell lysis by ultrasonication, French pressure cell, Centrifugation and Cell free supernatant, Product extraction by protein separation techniques, Protein purification by ultrafiltration and chromatographic separation, and freedrying to be explained.

11. **State the sensing and monitoring principles of Pressure, dO_2 and dCO_2 measurements. Nov 2014**

    Pressure measurement: Bourdon tube pressure gauge with neat diagram, Diaphragm type pressure sensor used, Agitator shaft power by watt meter and rate of stirring by tachometer, Foam measurement and control by simple circuit loop. The importance of controlling dissolved oxygen, and Galvanic and Polarographic type of probes and its material of construction with electrolyte assembly. The concept of pH probe used in dissolved carbondioxide measurement to be explained with a measure of one unit pH change when 10 folds of dCO_2 raise to be stated.

12. **Write a detailed note on material of construction of fermentor (Nov / Dec 2011)**

    Material of construction should withstand pressure, Sterilization, Corrosion, potential toxicity, cost etc. AISI standard, Stainless steel composition, areas where wood, plastic and concrete used, corrosion resistance, thin hydrous oxide film on the surface of the metal, stabilized by chromium and non-porous, insoluble self healing material. Increased chromium content increases resistance, presence of molybdenum increases resistance of SS to halogen salts and seawater- electro polishing done, and Thickness of construction varies with size of the fermentor

13. **Discuss about the different agitators and valves of fermentation system (April /May 2012)**

    Draw different types of Agitators, and concept of vortex formation and avoidance, and different valves system, Aseptic requirement of fermentation- joints of glass to glass, glass to metal and metal to metal, glass to metal joins with gasket, lip seal and O ring type diagram and explain each, Nitryl /butyl rubbers used for seal, double O ring with steam lock according to level of operations. Explain the principle and operations of stuffing box packed gland seal, simple bush seal, and mechanical seal and magnetic drives; discuss its advantages and disadvantages. The ancillary equipments, such as valves and its types to be mentioned with neat diagrams. The concept of coarse adjustment and fine adjustment and fitness for sterile usage etc. The types such as Gate valve, Globe valve, Piston valve, Needle valve, Plug valve, Ball valve, Butterfly valve, Pinch valve, Diaphragm valves to be explained with its fitness for sterile use and course or fine adjustments etc.

14. **With a neat diagram, explain the configuration of fermentor and its ancillaries (Nov / Dec 2012)**
15. **Compare and explain on-line and off-line control (Nov/Dec 2014)**
16. **Discuss about the different process parameters and its control (April /May 2014)**

Unit II FERMENTATION PROCESS

PART – A QUESTIONS AND ANSWERS

1. State few characteristics of Ideal Production medium

Maximum yield of product, maximum rate of product formation, minimum yield of undesired products, less problem in down stream operations.

2. Mention the elemental composition of Bacterial cell.

Carbon : 50-53 % dry weight, Nitrogen: 12-15%, Hydrogen: 7%, Phosphorus: 2-3%, Sulphur: 0.2-1.0 %, Sodium&Calcium: 0.5-1.0%, Pottasium:1.0-4.5%

3. Write the properties of 'Ideal Antifoam'.

Antifoam should readily reduce the surface tension so that foaming can be suppressed, also it should not be assimilated by the organism and should be easily removed during downstream processing.

4. Write a note on addition of Precursors in the medium.

Precursors: are the chemical added to incorporate in to the product formed, They become part and parcel of the product. eg. CSL added to penG production increases 20 units to 100 units/cm3, CSL found to be contain phenylethylamin, which seeded in PenG molecules to yield more.

5. Explain about Serum based media.

Used for animal cell cultivation contains ~ 1000 components such as organic, inorganic salts, amino acids, vitamins, carbon sources, hormones, growth factors, hemoglobin, albumin, etc. Most of the3m are not needed for cell growth and differentiation of the cells. Serum based medium should be free from bacterial, viral or pyrogen free contamination. Eg. Foetal cvalf serum.

6. Mention the commonly used carbon sources.

Cane molasses, Beat molasses, Corn & Maize starch, Oils, fats, hydrocarbons, Whey.

7. Mention commonly used Nitrogen sources.

Inorganic forms : Ammonia, Nitrate salts etc Organic forms: Protein, Peptide and amino acid, urea. Complex media such as Corn steep liquor, Soya bean meal, Pharma media, Yeast extract.

8. What are the nutritional requirements for the growth of Microorganism?

Medium should contain Carbohydrates, Nitrogen, Minerals such as S, P, Mg, K, Fe, Zn, Cu, Mn, Chelators, growth factors, Precursors, inhibitors, inducers, Oxygen requirements for fast metabolism and better rheology, antifoams.

9. What is meant by Defined media?

Medium with known elemental composition and its concentration

10.State the Stoichiometry for growth and product formation.

Carbon and Energy sources + Nitrogen sources + O_2 + other requirements ---- Biomass + Products + CO_2+H_2O+ heat

11.What is meant by yield coefficient (Y)?

Y = Qnty. Of cell dry matter product/ Qnty of carbon substrate utilized

12. What is aim of Plackett-Burman design?

When more than five independent variables are used investigated, to find the most important variable in system, then can be optimized for further studies.

13.Mention any four criteria for large-scale production medium. (Nov / Dec 2012)

Maximum yield of product, maximum rate of product formation, minimum yield of undesired products, less problem in down stream operations.

14. Mention the types of Response surfaces outputs in medium design.

Mound, Rising ridge, Saddle etc

15. Write about Yeast extract.

Backers yeast undergoes autolysis at 50-55 deg C, Plasmolysis with high concentration of NaCl that releases aminoacids, peptides, vitamins etc

16. Mention the elemental composition of Bacterial cell.

Carbon: 50-53, Nitrogen: 12-15, Hydrogen: 7, Phosphorus: 2-3, sulphur: 0.2-1.0, Pottassium; 1-4.5 % of dry weight.

17. Write a short note on Minimal medium.

In aerobic fermentation of microbial cell, should have minimum quantities, for economic design of media if component wastage is to be minimal. These minimal quantities of nutrients should be calculated for specific production of biomass and product without wasting.

18. Write the composition of Cane Molasses?

Water: 20%, ash: 8%, Total sugar: 40-60%, Total Nitrogen: 3%, Gums: 2%, Freeacids:2%

19. What is the necessity of adding Inducer in the medium?

Inducers are added to form certain specific products and also alters the cellwall permeability eg. glyceral

20. Explain Sulphite waste liquor and its uses.

The wood chips are hydrolysed by calcium bisulphate under high temperature and pressure, the spent liquior has 10-12% of solids of cellulose and 2% of sugar of hexoses and pentose, that is used in ethanol production.

PART B QUESTIONS

1. Give a detailed account on various Carbohydrate sources used in media design

Carbon source and importance, types of sources such as natural sources, Oils / fats, Hydrocarbons, etc, It decides the rate of process and product cost. eg, starch extracted from Maize grain or potatoes, starch is readily hydrolyzed by dilute acids and enzymes to give a variety of glucose syrups. Sucrose is obtained from sugarcane, beetroot etc.

The rate at which C source is metabolized can often influence the formation of biomass / Production of primary / secondary metabolites. Fast growth due to rapid utilization of sugar to decrease the production of secondary metabolites experimentally proved by glucose usage and lactose usage. Other carbohydrate sources used such as Oils and Fats, Molasses, cheese whey, cereals, sulphite waste liquor, rice straw, malt extract etc.

2. Explain the different Nitrogen sources used in medium design

Nitrogen sources such as inorganic and organic sources, energy supplements from nitrogen sources etc to be explained. Inorganic: Ammonia gas, ammonium salts and nitrates, pH control in defined medium. Ammonium salts, nitrates used the free acid will be liberated pH altered.

Nitrate reeducates is repressed by ammonia till its presence in medium, once diminished then nitrate assimilated. Hence mixture of two nitrogen sources regulates other's intake.

Organic : Supplied as amino acid, protein, urea, growth will be faster with a supply of organic nitrogen. Proteinecious nitrogen compounds forms as source for amino acids like corn steep liquor, soya bean meal, peanut meal, cotton seed meal, yeast extract. But storage affect by moisture, temperature.

3. Write a detailed account on Plackett-Burman design of media optimization.

The principle of Medium optimization, careful consideration of each elemental composition for efficient grown biomass and high productivity, method to use of trial and error method, time consuming with X^n where x is no. Of levels and n is no. of variables. Plankett Burman design used when more than five independent variable are used, most important variable in a system can be found and optimized further. Technique allows use of X no. of experiments with X-1 variable and X must be 4 or multiples of 4, concept of dummy variable to be introduced, example chart to be drawn with high and low concentration of each variable. As the analysis for each variables the Difference to be found, Effect of each variable, mean square of variable to found, procedure of experimental error and finally the factor of each variable to be seen using F test. These values to see for the effect of each variable in the growth of microorganism.

4. Explain the various Response surface methods of optimization: (April / May 2011)

Three factors are taken for experiment and responses surface of contour plots or Topographical maps with constant elevation values. Response means the result of an experiment carried out at particular value of variable being investigated. To construct a contour plot, the responses of series of experiment, employing different conditions combinations are inserted on the surface of plot. The points giving same results are then joined together to make contour line, Strategy to arrive a optimum to made as 5 steps and further optimization can be done. Explain the types of Mound, Saddle, Rising ridge etc. The significance of Simplex method: two axis and experimental movement – Centroid and Crabwise way movement to meet the optimum etc.

5. Write notes on Precursors and Inducers in Medium formulation.

Precursors: are the chemical added to incorporated to the product formed eg. CSL added to penG production increases 20 units to 100 units/cm3, CSL found to be contain phenylethylamine, which seeded in PenG molecules to yield more.

Inducers: Chemical that are important, that only in the presence of such chemicals the product would be formed. Eq. Substrates or substrate analogues. All enzymes are inducible eg. Starch for amylase, Maltose for pullulanase, Pectin for pectinase, fatty acids for lipase etc.

6. Explain Plackett Burman and Reponse surface methods in the optimization (Nov / Dec 2013)

Three factors are taken for experiment and responses surface of contour plots or Topographical maps with constant elevation values. Response means the result of an experiment carried out at particular value of

variable being investigated. To construct a contour plot, the responses of series of experiment, employing different conditions combinations are inserted on the surface of plot. The points giving same results are then joined together to make contour line, Strategy to arrive a optimum to made as 5 steps and further optimization can be done. Explain the types of Mound, Saddle, Rising ridge etc.

7. How will you formulate medium for optimal growth (Nov/Dec 2013)

8. Explain the advantages and considerations of Animal Cell culturing media

9. What are the criteria for an ideal production medium

10. Explain the process of medium formulations (April/May 2012)

11. What are the Minerals and Growth factors required for cell culture (April/May 2012)

12. What are the different precursors and its importance

13. What is Oxygen requirement and related factors

14. Explain the different plots of medium optimization.

15. Discuss serum based medium and bioreactor design for Animal cell cultivation.

Unit III MEDIA DESIGN AND STERILIZATION FOR FERMENTATION PROCESSES

PART – A QUESTIONS AND ANSWERS

1. List down various methods of Sterilization.

Steam, Dry heat, Pasteurization, Tyndallisation, Chemical, Ionizing radiation, filtration

2. Define Del factor

Measures fractional reduction in viable organism count produced by certain heat and time regime (degree of sterilization)

3. What are the advantages of filter sterilization?

Physical exclusion of microbes from air and liquid medium- thermal liable solutions can be sterilized.

4. State the need of sterilization in fermentation with two reasons.

Medium would have to support the growth of production organism and also contaminant cause less productivity. If fermentation is continuous, contaminates may outgrow the other and displace it from system

5. Explain the concept of HTST in sterilization.

The concept of high temperature and short time is the basis of continuous sterilization and same del factor can be attained.

6. List the advantages of 'Continuous sterilization'.

Advantages as no contact of steam medium. No sedimentation so the process gets over quickly – disadvantages as Foaming may occur during heating.

7. Calculate the Del overall for the steam sterilization using logical data.

Consider the No as 10^{11} cell and Nt value as 10^{-3} and calculate del as ln(No/Nt)

8. State and explain the log penetration equation.

ln(Nt/No) = -kx where Nt & No concentration of particle in the air before and after filtration; x is length of filter; k is constant.

9. State the Activation energy and Arrrhenius constant for B. Stearothermophillus.

Activation energy (E) = 67.7 kcal/mole; Arrhenius constant = $1x10^{36.2}$ / sec

10. Explain the concept of 'Cooking effect' in medium sterilization

During sterilization, the initial rise of yield is due to some nutrient made more available, thi effect is called cooking effect.

11. State the process of tyndallization

Devised by Tyndall 1660, spontaneous generation, heating the material at atmospheric pressure to 100 deg C on successive days, useful in medium gets affected more than 100 deg C, where spores are present. These spores which survive the first heat treatment germinate into vegetative cells which are destroyed in the second heat generation.

12. State the uses of Dry heat of sterilization

Used in Incubators and ovens, used to sterilize glassware and heat stable solid materials, require much longer exposure as heat conduction is less rapid in air (poor thermal conductivity) than in steam spore become more heat resistant in the absence of moisture.

13. State the Mechanism of Ionizing radiation

The use of high energy electrons, UV rays, gamma rays used for the media and air sterilization, not fully reliable but highly used in plastic lab ware, petri-dishes. UV rays are effective in killing air borne microorganism.

14. What are the types of Filters used in Sterilization?

Depth filters: Variety of natural fibers such as cotton replaced by glass fibre (glasswool) to form as bed ensures removal of organisms.

Exclusion filters: The PTFE membranes with pore size 0.2-0.45um for all organisms to remove used. These filters are used in heat labile liquid sterilization. These are hydrophobic membranes cartridges for air filtration, they are unidirectional.

15. List down the mechanisms by which filters sterilization by membranes

Inertial impaction, Diffusion, Electro-static attraction and interception etc

16. Explain the phenomena of Interception of membrane filtration

The fibers of filter interwoven to define size of the openings, The particle which are large to these pores removed by direct interception. Even small particles also getting removed by interception because more than one particles arrive at pore simultaneously and irregular particle tend to bridge a pre which are much small still to filter small particles.

17. State the equation of del Holding calculation.

Del holding = Del overall - Del heating – Del cooling

18. State the mechanism of Continuous sterilization

Temp of the medium elevated in continuous heat exchanger and maintained in insulated serpentine holding coil for the holding period and cooled using flash coolers.

19. State the advantages of Spiral heat exchangers in Continuous sterilizer

Two streams are separated and no contact and no cross contamination.

This aids in self cleaning of suspended solids, sedimentation because of spiral route of flow

20. State the need of empty vessel sterilization

In order to minimized the time for sterilization and overall duration of the fermentation, medium is sterilized separately and the empty fermentation vessel is heated with heating coil and jacketed steam supply and after 20 minutes of sterilization, pre-sterilized medium is added with positive pressure.

Part B QUESTIONS

1. Explain the various methods of sterilization process.

Dry heat: used in incubators / Ovens – used to sterilize glassware / heat stable solid materials – requires much longer exposure as heat conduction is lass rapid in air because of poor thermal conductivity than in steam - Spore become more heat resistant in absence of moisture.

Pasteurization: It is not strictly a method of sterilization; it is method of making food and beverages for safe consumption. Techniques involved in heating the substance up to its boiling point for a short period, which is sufficient to kill many organisms, eg. Milk at 62 deg C for 30 minutes.

Tyndalization: devised by Tyndall 1660- spontaneous generation. Heating the material at atmospheric pressure to 100 deg C on successive days.- Useful in medium gets affected more than 100 deg C temperature, where spores are present – Those spores which survive the first heat treatment germinate into vegetative cells which are destroyed in the second heat generation.

Ionizing radiation: High energy electrons, UV rays gamma rays used for media and air sterilization, Not fully reliable- Radiation used in plastic lab ware, petridishes etc, UV lamps installed in culture or clean room, effective in killing air borne microorganisms.

Chemical are not used in medium sterilization, only used in reactor washing. But used to disinfect apparatus, glassware etc. Used to clean pipettes, chemical sterilent is to be toxic volatile so that it can be removed easily. Ethylene oxide is best, not widely used because it is used as explosive. Other examples of chemical sterilents are Formaldehyde, phenolics such as Lysol.

2. Explain in detail the design and scale up of Batch sterilization process

The complete derivation procedure as per the class notes, related to batch sterilization process, where to mention the points like first order reaction of destruction of micro organism, explain V, N, t, K term separately, derive and show the Arrhenius plot and calculation of activation energy and prediction of reaction rate for any temp, define del factor and derive, state its importance and D&H plot to be shown.

3. Give a detailed account on design and scale up of Continuous sterilization process.

State the same del factor can be achieved by increasing the temperature and degreasing the time of exposure – HTST method only possible rapid heating and rapid cooling – using spiral heat exchangers (double spiral type) – two sheets of high grade Stainless steel curved around central axis to form double spiral setup – steam allowed in one spiral and medium allowed in another spiral – used as countercurrent principle of direction used for heating or cooling liquid. The design aspects should also be included as per Stanberry book. Advantages as no contact of steam medium. No sedimentation so the process gets over quickly – disadvantages as Foaming during heating.

4. Derive the Sterilization kinetics and explain its significance.

The complete derivation procedure as per the class notes, related to batch sterilization process, where to mention the points like first order reaction of destruction of micro organism, explain V, N, t, K term separately, derive and show the Arrhenius plot and calculation of activation energy and prediction of reaction rate for any temp, define del factor and derive, state its importance and D&H plot to be shown The complete derivation procedure as per the class notes, related to batch sterilization process, where to mention the points like first order reaction of destruction of micro organism, explain V, N, t, K term separately, derive and show the Arrhenius plot and calculation of activation energy and prediction of reaction rate for any temp, Del factor and its importance. Deindoerfer and Humphrey 1959 used the term ln[N0/Nt] as design criterion for sterilization, which has been called as “Del factor” or Nabla factor or Sterilization criterion. It measures fractional reduction

in viable organism count produced by certain heat and time regime (degree of sterilization) (Del = $A.t.e^{-E/RT}$), state its importance and D&H plot etc

5. Explain the significance of Sterilization and various consequences of contamination in Bioprocess. (April/May 2009)

Medium would have to support the growth of production organism and also contaminant cause less productivity. If fermentation is continuous, contaminates may outgrow the other and displace it from system. The foreign organism may contaminate the final product. The contaminant might also produce product, hence difficult to extract other. The contaminant would degrade the desired product. Phage contamination in bacterial cells may cause cell lysis.

6. A continuous culture system is being contracted. The fermentation tank is to be 50000L in size and the residence time is to be 2 hrs. A continuous sterilizer to be used. The unsterilized medium contains 10^4 spores/L. The value of Kd has been determined to tbe 1 min-1 at 121 deg C and 61 min-1 at 140 deg . For each temperatures (121 deg C and 140 deg C) determine the required residence time in the holding section so as to ensure that 99% of the time in four weeks of continous operation can be obtained without contamination. (Nov / Dec 2012)

7. It is required to provide a $20m^3$ fermentor with air at a rate of $10m^3$/min for a fermentation lasting 100 hrs. From an investigation of filter material to be used the optimum linear air velocity was shown to be 0.15m/sec at which the k value was 1.5351/cm. The air in the fermentation plant contained approx. 200 microbes/m^3. Calculate the dimension of the filter. Assume Standard acceptable degree of contamination i.e. 1 in 1000.

N_o = Total amount of air provided x 200 , N_o = 10x60x100x200 = 12 x 10^6

Nt = acceptable degree of contamination Nt = 10^{-3}

ln $\{10^{-3}/(12x10^6)\}$ = -kX ; ln 8.33 x 10^{-11} = -1.535X; X (length) = -23.21/-1.535 = 15.12 cm, Cross section of filter is volumetric air flow rate divided linear air velocity Πr^2 = 10/0.15 x 60: r (radium) = 0.59m.

8. Explain the usage of 'Chemical sterilents' in Bioprocess technology. (April/May 2010)

Chemical are not used in medium sterilization, only used in reactor washing. But used to disinfect apparatus, glassware etc. Used to clean pipettes, chemical sterilent is to be toxic volatile so that it can be removed easily. Ethylene oxide is best, not widely used because it is used as explosive. Other examples of chemical sterilents are Formaldehyde, phenolics such as Lysol.

9. Explain the sterilization process using Pasteurization, Tyndalization and Ionizing radiation

Pasteurization: It is not strictly a method of sterilization; it is method of making food and beverages for safe consumption. Techniques involved in heating the substance up to its boiling point for a short period, which is sufficient to kill many organisms, eg. Milk at 62 deg C for 30 minutes. Tyndalization: devised by Tyndall 1660- spontaneous generation. Heating the material at atmospheric pressure to 100 deg C on successive days.- Useful in medium gets affected more than 100 deg C temperature, where spores are present – Those spores which survive the first heat treatment germinate into vegetative cells which are destroyed in the second heat generation.

Ionizing radiation: High energy electrons, UV rays gamma rays used for media and air sterilization, Not fully reliable- Radiation used in plastic lab ware, petridishes etc, UV lamps installed in culture or clean room, effective in killing air borne microorganisms.

10. Explain in details, how you will design Batch sterilization process. (April/May 2010)

The process sterilization, Heating Holding Cooling cycles, Temperature and time profile as per the class notes, Del overall calculation with design organisms, and Relationship of Del with temp and time profile using Richard's graphical method, Calculation of Holding time with specific death rate of *Bacillus Sterothermophillus*. Scale up of batch sterilization on volume to be given

11. Write a detailed note on Filter sterilization, types and uses. (Nov 2013)

Physical exclusion of microorganisms for air and medium sterilization. Depth filters, Exclusion filters – Variety of natural fibers such as Cotton – replaced by glass fiber to form as bed ensures removal of organisms – sterilized by steam also – but concentration of contaminants increased and pressure drop seen – efficiency of filtration to be decreased. Used in input or exhaust air prefilter purpose. Exclusion filters are modern cartridges used membranes with pore size of 0.2 – 0.45micrometer for all organisms to remove used, heat labile liquid sterilization, antibiotics used. Large hydrophobic membrane used unidirectional steam flow used. State the table of differences of Depth and membrane filters

12. Discuss the various measures by which Contamination can be prevented in Bioprocess.

Avoidance of contamination steps such as using pure inoculums to start the fermentation – sterilizing the medium to be employed – sterilizing the fermentor vessels – sterilizing all material to be added to the

fermentation process – maintaining aseptic conditions during the fermentation – Fermentation by protected medium – less pH, only some can grow etc to be explained in detail.

13. Derive the Sterilization kinetics and explain its significance. (Nov/Dec 2013)

The complete derivation procedure as per the class notes, related to batch sterilization process, where to mention the points like first order reaction of destruction of micro organism, explain V, N, t, K term separately, derive and show the Arrhenius plot and calculation of activation energy and prediction of reaction rate for any temp, define del factor and derive, state its importance and D&H plot to be shown The complete derivation procedure as per the class notes, related to batch sterilization process, where to mention the points like first order reaction of destruction of micro organism, explain V, N, t, K term separately, derive and show the Arrhenius plot and calculation of activation energy and prediction of reaction rate for any temp, Del factor and its importance. Deindoerfer and Humphrey 1959 used the term ln[N0/Nt] as design criterion for sterilization, which has been called as "Del factor" or Nabla factor or Sterilization criterion. It measures fractional reduction in viable organism count produced by certain heat and time regime (degree of sterilization) (Del = $A.t.e^{-E/RT}$), state its importance and D&H plot etc

14. How will you design Continuous Sterilization process (Nov/Dec 2013)

State the same del factor can be achieved by increasing the temperature and degreasing the time of exposure – HTST method only possible rapid heating and rapid cooling – using spiral heat exchangers (double spiral type) – two sheets of high grade Stainless steel curved around central axis to form double spiral setup – steam allowed in one spiral and medium allowed in another spiral – used as countercurrent principle of direction used for heating or cooling liquid. The design aspects should also be included as per Stanberry book. Advantages as no contact of steam medium. No sedimentation so the process gets over quickly – disadvantages as Foaming during heating.

17. How would you design of sterilization equipments? (April/May 2012)

18. Explain the sterilization methods used for air and liquid media (Nov / Dec 2012)

19. Explain the thermal death kinetics of microorganisms (Nov / Dec 2012) (Nov/Dec 2013)

Unit IV METABOLIC STOICHIOMETRY AND ENERGETICS

PART – A QUESTIONS AND ANSWERS

1. What is meant by Electron balance?

The number of electrons available for transfer to oxygen on combustion of substance to CO2, H2O and Nitrogen containing compounds.

2. Define Degree of reduction

The number of equivalents of available electrons in 1 g of atom carbon

3. Define 'System' and 'Surrounding' as Thermodynamic point of view.

System: consists of any matter identified for investigation (cell)

Surrounding: is the environment of system lies and set apart from it (medium)

4. State and explain briefly the Steady state condition.

Continuous process: allows matter to flow in and out of system – if rates of mass input and output are equal – can be operated indefinitely. If all the properties of a system such as temperature, pressure concentration, volume, mass do not vary with time called Steady state.

5. State the General Mass Balance Equation.

Mass accumulated = In system	mass in through system boundaries	-	Mass out thro. System Boundary	+ Mass Generated by system	- Mass consumed by system

6. Write Elemental equation for Microbial aerobic growth using Carbon, Nitrogen and O_2.

$C_wH_xO_yN_z + aO_2 + bH_gO_hN_i$ --- $cCH_\alpha O_\beta N_\gamma + dCO_2 + eH_2O$

7. Calculate Degree of reduction for Glucose

$C_6H_{12}O_6 = 6(4) + 12(1) + 6(-2) = 24$; $\gamma = 24/6 = 4$

8. How will you determine the microbial reaction is exo / endothermic.

Heat of the reaction can be determined and estimated using microbial stoichiometry and the concept of available electron. $\triangle h_c$ value is calculated for standard heat of reaction, $\triangle H_{rxn} = \triangle h_{c(reactant)} - \triangle h_{c\ (prduct)}$

If the result is positive the reaction is endothermic and heat is absorbed or vice versa

9. What are the types of material balances?

Differential balance method, Integral balance method

10.Write a short note on Theoretical Oxygen Demand.

Oxygen requirement is directly related to the electrons available for transfer to Oxygen. Hence O_2 demand derived from electron balance.

11. Define 'Respiratory Coefficient'.

It is the ratio of moles of CO_2 produced with moles of O_2 consumed

12. Define 'Open System' and 'Closed System' as Thermodynamic point of view.

Open system : If the system boundary allows masses to flow in and out of the system

Closed system : If the system boundary allows masses to flow in and out of the system

13. Define Yield Coefficient of biomass yield

It is the ratio of the quantity of biomass produced using quantity of carbon sources used

14. Explain Continuous process on Thermodynamic point of view

It is a open system that allows matter to flow in and out of system. If rates of mass input and out put are equals, then continuous processes can be operated indefinitely.

15. What is the equilibrium concept as thermodynamic point of view

In a system if all the opposing forces are exactly counter balanced, so that the properties of all system do not change with time. Then it is under equilibrium condition. System tend to approach equilibrium, when they are isolated from their surrounding, where there is no net change in system/surround, the energy of the system is minimum or static

16. State the General Steady state mass balance equation

Mass in + mass generated = mass out + mass consumed

17. What is meant by Engineering judgment in Material balance calculations

Stating all assumptions applied to the problem is called Engineering judgment. Real time situations are complex, so one or more assumptions are required for calculations. The omitted details are assumed, provided assumptions are reasonable. These assumptions to be stated exactly to reveal other, so that it can be improved or accepted.

18. Calculate the degree of reduction of Ethanol

$C_2H_5OH = 2(4) + 6(1)+1(-2) = 12$; $\gamma = 12/2 = 6$

19. State the Product yield in a biological reaction.

It is the ratio between gms of product formed / gms substrate consumed

20. State the equation for heat of the microbial reaction

Heat of the reaction can be determined and estimated using microbial stoichiometry and the concept of available electron. $\triangle h_c = -q\ \gamma x_c$; where $\triangle h_c$ = Molar heat of combustion, q= heat evolved per mole of available electron transferred O_2 during combustion. γ = degree of reduction of defined compound and x_c= nol of carbon atoms in molecular formula.

Part B QUESTIONS

1. Explain in detail about the types and procedure of 'Material balancing'.

Differential balance and Integral balance methods to be explained with principles, First step is to understand the problem, with information available, by using best standard procedures, the unknown quantities should be calculated

(1) Draw a clear process flow diagram showing all relevant information with simple box diagram with given variable masses with molar quantities.

(2) Select a set of units, to make calculation much easier if all the values expressed in same types of units.

(3) Select a basis of calculation: to visualize the problem, focus on single quantity of material entering or leaving system

(4) State all assumption applied to the problem with engineering judgment, omitted detail is assumed

(5) Identify the particular component of the system involved in reaction then assemble, analyze, calculate and finalize the problem

2. Explain the following with suitable examples (i) System (ii) boundary (iii) Steady state (iv) Equilibrium

Thermodynamics are science dealing with properties of matter – useful in setting up material balances. System: consists of any matter identified for investigation (cell) Surrounding: is the environment of system lies and set apart from it. System boundary: system remain isolation from surrounding by system boundary- eg. Walls of the beaker or fermentor or any imaginary boundary – if boundary does not allow mass to pass from system to surrounding called Closed system – If boundary exchanges mass from to system and surrounding called Open system - Continuous process: allows matter to flow in and out of system – if rates of mass input and output are equal – can be operated indefinitely. If all the properties of a system such as temperature, pressure concentration, volume, mass do not vary with time called Steady state. If all the opposing forces are exactly counter balanced so the properties of the system do not change with time called equilibrium.

3. Discuss the Batch process, Fed batch process, and Continuous process with Thermodynamics point of view

Thermodynamics are science dealing with properties of matter – useful in setting up material balances. System: consists of any matter identified for investigation (cell) Surrounding: is the environment of system lies and set apart from it. System boundary: system remain isolation from surrounding by system boundary- eg. Walls of the beaker or fermentor or any imaginary boundary – if boundary does not allow mass to pass from system to surrounding called Closed system – If boundary exchanges mass from to system and surrounding called Open system

Batch process: operates in closed system. All materials are added to the system at the start of the process, The system is the closed and products removed only when the process is complete. Fed batch process: a type of semi batch process – allows only input of materials to the system but not output. Continuous process: allows matter to flow in and out of system – if rates of mass input and output are equal – can be operated indefinitely.

4. Derive and discuss growth stoichiometry & Elemental balance for biomass production.

Despite of thousands of intracellular reactions, the cell growth obeys the law of conservation of matter. Carbon, hydrogen, oxygen, nitrogen are consumed during rowth to form new cells. We consider 2 extracellular products apart from new cell formation as CO_2 and H^2O.

For aerobic growth we write the eqn. as

$CwHxOyNz + aO_2 + bHgOhNi - cCH_\alpha O_\beta N_\delta + dCO_2 + eH_2O$.

CwHxOyNz = formula of substrate (for glucose w=6, x=12, y=6, z=0)

bHgOhNi = formula for Nitrogen source $cCH_\alpha O_\beta N_\delta$ = formula for dry biomass, where a,b,c,d,e are stoichiometric coefficients. The mass balance can be done (stoch. Coeff X no. of atoms = atomic balance) C balance is w=c+d ; H balance is $x+bg = c_\alpha + 2e$; O balance is $y+2a+bh = c_\beta +2d+e$;
N balance is $z+bi = c_\delta$

5. **Production of Single cell protein from hexadecane is described by the following reaction equation: $C_{16}H_{34} + aO_2+bNH_3$ -----$cCH_{1.66}O_{0.27}N_{0.20} +dCO_2 +eH_2O$ Where $CH_{1.66}O_{0.27}N_{0.20}$ represents the biomass. If RQ is 0.43, determine the Stoichiometry coefficients. (April/May 2009)**

Answers as per the Solved example 4.7 about Stoichiometry coefficients for cell growth page no 76 of Pauline and Doran book.

6. **Explain in detail about the types and procedure of 'Material balancing'. (April / May 2011)**

Differential balance and Integral balance methods to be explained with principles, First step is to understand the problem, with information available, by using best standard procedures, the unknown quantities should be calculated 1. Draw a clear process flow diagram showing all relevant information with simple box diagram with given variable masses with molar quantities. 2. Select a set of units, to make calculation much easier if all the values expressed in same types of units. 3. Select a basis of calculation: to visualize the problem, focus on single quantity of material entering or leaving system 4. State all assumption applied to the problem with engineering judgment, omitted detail is assumed 5. Identify the particular component of the system involved in reaction then assemble, analyze, calculate and finalize the problem

7. **(i) Humid air enriched with O_2 is prepared for gluconic acid fermentation. The air is prepared in special humidifying chamber. 1.5 lit/h of water enters the chamber at the same time as dry air and 15g mol/min dry O_2. All the water is evaporated. The out flowing gas is found to contain 1% w/w water. Draw and label the flow sheet for this process.**

   (ii) Derive and discuss growth stoichiometry and detailed elemental balance for biomass production. (April / May 2010)

(i) Let us choose units of 'g' and 'min' for the whole process. The information first converted to mass flow rates in these units. The density of water is taken to be 10^3 g/lit. =1.5 lit/h = 1.5 lit/h (10^3g/l)(1hr/60min) = 25g/min Molecular weight of O_2 is 32. = 15 gmol/min = 15gmol/min (32g/gmol) = 480 g/min, Draw the flow sheet as per the class notes.

(ii) Despite of thousands of intracellular reactions, the cell growth obeys the law of conservation of matter. Carbon, hydrogen, oxygen, nitrogen are consumed during rowth to form new cells. We consider 2 extracellular products apart from new cell formation as CO_2 and H^2O. For aerobic growth we write the eqn. as $CwHxOyNz + aO_2 +bHgOhNi – cCH_\alpha O_\beta N_\delta + dCO_2 +eH_2O$. derivation to be done. Up to finding 5 coefficients from 4 balance equations.

8. **Acetobacter aceti bacteria convert ethanol to acetic acid under aerobic conditions. A continuous fermentation process for vinegar production is proposed using viable A. aceti cells immobilized on the surface of gelatin beads. The production target is 2kg h-1 acetic acid; however the maximum acetic acid concentration tolerated by the cells is 12%. Air is pumped into the fermentor at a rate of 200 g mol per hour. Calculate**
   1. What minimum amount of ethanol is required?
   2. What minimum amount of water must be used to dilute the ethanol to avoid acid inhibition?
   3. What is the composition of the fermentor off gas?

9. **(i) State the 'Law of Conservation of Mass' and prove that Mass in = Mass out.**

Explanations and assumptions, - The General Mass balance equation and types of material balances like Differential balance and Integral balance- Simplification of General Mass balance equation and finally the steps involved in Mass balance calculations. State the following in full explanation as per the class notes date – Law of conservation of mass

   (ii) A continuous process is set for the Treatment of wastewater. Each day 10^5 kg of cellulose and 10^3 kg of bacteria enter in the feed stream, while 10^4 kg of cellulose and $1.5x10^4$ kg of bacteria leave in the effluent. The rate of cellulose digestion by bacteria is $7x10^4$ kg/day; the rate of bacterial growth is $2x10^4$ kg/day. The rate of cell death by lysis is $5x10^2$kg/day. Write balances for cellulose and bacteria in the system. (April/May 2010)

Cellulose is not generated by the process, only consumed hence
For Cellulose (10^5-10^4 + 0- $7x10^4$) = $2x10^4$ kg of cellulose accumulation/day.
For Bacteria: (10^3 - $1.5x10^4$+ $2x10^4$ -$5x10^2$) = $5.5x10^3$kg/day of bacteria.

10. **(i) Write a note on theoretical O_2 demand in metabolic stoichiometry.**

    (ii) Fumaric acid is produced from malic acid using the enzyme, fumarase. Calculate the standard heat of reaction for the following enzyme transformation: $C_4H_6O_5$ ------ $C_4H_4O_4 + H_2O$ Standard heat of combustion of Maleic acid is –1328.8 KJ/gmol & that of Fumaric acid is –1334 KJ/gmol. (April/May 2010)

11. **Xanthan gum is produced using Xanthomonas sp. in Batch culture. Laboratory experiments have shown that for each gram of glucose utilized by the bacteria, 0.23g oxygen and 0.01 g ammonia are**

consumed, while 0.75 g gum, 0.09g cells, 0.27 g gaseous CO_2 and 0.13 g H_2O are formed. Medium containing glucose and ammonia dissolved in 20000 liters water is pumped into a stirred fermentor and inoculated with culture. Air is spurged into the fermentor; the total amount of off-gas recovered during the entire batch culture is 1250 kg. Because of final high viscosity, the gum should not be allowed to exceed 3.5wt% A. How much glucose and ammonia are required? B. What percentage excess air is provided?

12. **Corn steep liquor contains 2.5 % invert sugars and 50% waste, the rest can be considered solid. Beet molasses containing 50% sucrose. 1% invert sugars, 18% waste and the remainder solids, is mixed with corn-steep liquor in a mixing tank. Water is added to produce a diluted sugar mixture containing 2%(w/w) invert sugars. 125 kg corn steep liquor and 45 kg molasses are fed into the tank. A. How much water is required? B. What is the conc. of sucrose in the final mixture?**
13. **Explain in detail about the types and procedure of Material balancing. (April/May 2011)**
14. **Explain heat generation by microbial growth (April/May 2012)**
15. **Explain the thermodynamic efficiency of growth. (Nov /Dec 2011)**

Heat of the reaction can be determined and estimated using microbial stoichiometry and the concept of available electron. It is found that such heat energy is related to degree of reduction as follows; $\triangle h_c = -qrx_c$ Where $\triangle h_c$ is molar heat of combustion, q is heat evolved per mole of available electron transferred O_2 during combustion r is degree of reduction defined compound, Xc is number of carbon atoms in molecular formula. $\triangle h_c$ value is calculated for standard heat of reaction, $\triangle H_{rxn} = \triangle h_{c(reactant)} - \triangle h_{c\ (prduct)}$ If the result is positive the reaction is endothermic and heat is absorbed or vice versa.

16. **Explain different models to predict specific growth rate. (April/May 2012)**
17. **Explain the models of cellular energetic and metabolism in aerobic growth of the Yeast *Saccaromyces cerevisiae.* (Nov / Dec 2012)**
18. **Discuss the advantages and disadvantages of batch, fed-batch and continuous cultivation of cells. (Nov / Dec 2012)**
19. **Aerobic degradation of acetic acid by microbes can be represented by following reaction: $C_6H_5COOH+aO_2+bNH_3$ --- $cC_5H_7NO_2+dH_2O+eCO_2$; Determine a,b,c,d and e if RQ=0.9 and Determine yield coefficients, $Y_{x/s}$ and $Y_{x/O2}$ and determine degree of reduction for the substrate and bacteria (Nov/Dec 2013)**
20. **Give detailed note on Growth and Product stoichiometry. (Nov / Dec 2011)**

    Despite of thousands of intracellular reactions, the cell growth obeys the law of conservation of matter. Carbon, hydrogen, oxygen, nitrogen are consumed during rowth to form new cells. We consider 2 extracellular products apart from new cell formation as CO_2 and H^2O. For aerobic growth we write the eqn. as $CwHxOyNz + aO_2 + bHgOhNi - cCH_\alpha O_\beta N_\delta + dCO_2 + eH_2O$. As per the class notes the further derivation to be done. Up to finding 5 coefficients from 4 balance equations. Another extracellular product $C_jH_kO_lN_m$ in the eqn. Hence it becomes $CwHxOyNz + aO_2 + bHgOhNi - cCH_\alpha O_\beta N_\delta + dCO_2 + eH_2O + f\ C_jH_kO_lN_m$ with this explain the Product yield coefficient Yp/s to be explained and theoretical product yield to be explained.
21. **Derive expression of Mass balance equation for a biochemical reaction. (April/May 2011)**

    State the law of conservation of Mass and explain the General mass balance equation and state Differential balance, and integral balance. Simplification of General mass balance equation and derive upto Massin= mass out- State the following in full explanation as per the class notes date – Law of conservation of mass

    Explanations and assumptions, - The General Mass balance equation and types of material balances like Differential balance and Integral balance- Simplification of General Mass balance equation (Mass accumulated= mass in –mass out + mass generated – mass consumed by the system) and finally the steps involved in Mass balance calculations such as continuous process at steady state, the accumulation term must be zero, the system at steady state cannot accumulates mass. Hence (Mass in + Mass generated = Mass out + mass consumed) The total mass neither be created nor destroyed except for nuclear reaction, so generation and consumption terms must be zero (Mass in = Mass out)

Unit V KINETICS OF MICROBIAL GROWTH AND PRODUCT FORMATION

PART – A QUESTIONS AND ANSWERS

1. Describe unstructured growth models.

These models view the cells as a single species in solution and attempt to describe the kinetics of cell growth based on cell and nutrient concentration profiles.

2. What are structured models?

Models, which incorporate the details of intracellular metabolism that views complex organization within the cell as Genetic system, Epigenetic system, metabolic system etc.

3. Explain Catabolism and Anabolism process

Catabolism: Energy containing molecules reduced to carbon-containing molecules are degraded to CO_2 and other oxidized end products, ATP etc.

Anabolism: Intermediates and end products formed from catabolism are incorporated into cell Constituents (DNA, RNA, lipids)

4. Explain the Segregated & Unsegregated view of modeling.

Segregated view = cell to cell heterogeneous consideration

Unsegregated view = Average cell approximation

5. Why specific growth rate / doubling time should be calculated at the log phase of the cell.

This is the phase where μ will be equal μmax hence it is right phase to calculate. The average time taken for the cell population to get doubled during the exponential phase of growth.

6. What is meant by 'Balanced growth' and Unbalanced growth

Balanced growth : When 'Specific growth rate' and 'Specific cell number growth rate' are equal, growth referred to be balanced, this is an ideal case.

Unbalanced growth: The variation in cell composition may occur even though cell number growth rate is constant, the cell mass growth rate will vary. This is the real time case.

7. State the significance of Plasmid instability.

The problem of plasmid loss – leads to large plasmid free population hence overall reactor efficiency decreases – Plasmid DNA mutation, segregation instability are the reasons – The total number of plasmids in generation must double in the next.

8. State and explain Luedeking & Piret Model

For lactic acid production by lactobacillus sp. Is a type of partially growth associated type where the production is found to depend on both concentration of cell as well as their growth rate. Hence the expression as $\gamma_p = \alpha\gamma_x + \beta x$; $\gamma_p = \alpha\mu x + \beta x$

9. What are the different phases of microbial batch growth? (April/May 2012)

Lag, Log phase, decline phase, stationary phase, death phase.

10. What are the types of product formations? (Nov / Dec 2012)

Growth associated product (Ethanol), Non growth associated Product (Vitamins, antibiotic), Partially growth associated product (amino acid, xanthan)

11. Describe the complex organization within the cell in Structural model.

Genetic system, Epigenetic system, metabolic system etc.

12. Write Plasmid Replication Model of Structured microbial growth kinetics

$1/(u_{pwo}-u) = K_p/V^{max}\ (1+K_p/P_s)$ where u_{pwo} = Plasmid washout growth rate; u = sp. Growth rate, K_p= Saturation constant, V^{max} = maximum rate of plasmid synthesis; P_s is steady state plasmid number.

13. Mention the reasons behind Death phase of the microbial growth pattern.

The death phase of the cell culture is due to cell losing viability or by lyses due to produced metabolites.

14. State the equations for specific growth rate and doubling time of the cell culture

$\ln X_t = \ln X_0 + ut$ where X_0 is the initial biomass concentration and u is the specific growth rate

Doubling time : ln2/u

15. Explain Monod model of microbial growth

It is assumed that only onne substrate is limiting the growth and cell proliferation

$U = u_{max}S/Ks+S$ where umax is the maximum specific growth rate of the cells, Ks is the value of the limiting nutrient concentration.

16. State the simple model for fraction of Plasmid bearing cells as a function of time

$\gamma\ x^+ = (1-P)\ u^+X^+$ where $\gamma\ x^+$ is the rate of growth of P+ cell population

17. Mention the growth model of filamentous organsim

Complicated multi-cellular and branched mould system that forms microbial pellets in the suspension. (dx/dt) = rho 4 pi r^2 where r is the radius of cell floc and rho is the density of the medium

18. State the models of Susbtrate inhibition of cell growth

Competitive inhibition : u= umS/Ks (1+s/ki)+S

Non competitive inhition : u= um/(1+ks/s)(1+s/ki)

19. State the models of Product inhibition of Cell growth

Uncompetitive product inhibition : u = umax / (Ks+s)(1+s/ki)

Competitive product inhibition : u = umS/Ks(1+p/Kp)+S

20. Mention Single cell model and explain

Reaction occurring in a single cell is being considered as the representative of the whole microbial population. For E.coli the DNA synthesis mass balance is written as $dM/dt = u(p/v/K_{mp}+p_3/v)\ (A/V/K_{ma}+A/V)F$ Where M = mass of DNA; u= rate constant for maximum rate of DNA formation per time; P= mass of deoxynucleotides; V= Cell volume. A= glucose conc. F= number of replication events.

PART B QUESTIONS

1. Explain the following (i)Unstructured growth & Unsegregate view (ii) Structured growth & segregate view

Phases of cell growth as the increase in cell number and cells to grow in size. Two phases involved in cell growth kinetics as Biological phase of cell and Environmental phase or growth medium. Single cell is a complicated multicomponent system and non homogeneous. Also the growing cell population show cell to cell heterogeneity. At a given point of time, in a given small region in space the cells vary with respect to age. These cells show different types of metabolic functions and activities. Hence by simplifying the complexicity of cell population kinetics we can propose assumptions like, a single component becomes the rate – limiting nutrient and others are present in high concentration. A inhibitory product which accumulates in the medium components influence in cell kinetics. Normally the external controls such as pH, temp, and dO2 concentrations are kept constant. Cellular components which are multicomponent are called Structured and single components representation is unstructured. When heterogeneous cells are considered it is segregated view point. When average cellular property is considered then it is unsegregated.

2. Write a detailed note on Unstructured growth model with reference to (i) Malthus model (ii) Monod model

Simplest models, view the cells as a single species in solution and attempt to describe the kinetics of cell growth based on cell and nutrient concentration profiles. These models does not account for dependency of exponential growth rate on nutrient conc. Instead they devised to have a maximum achiefvable population for a given nutrient. Explain the principles of Malthus model, Verlhust, pearl, reed model etc. Monad model: simplest model, includes the effect of nutrient conc. Based on observation of the growth of E.coli at various glucose conc. It is assumed that only one substrate is important in determining the rate of cell proliferation, similar to MM equation of enzyme kinetics, infact if substrate transport to the cell is limited by the activity of a enzyme permease. Hence cell growth follows as $\mu = u_{max}\ S/Ks + S$ and for batch growth at constant volume it dx/dt = umaxSX/ks+S where umax is the maximum specific growth rate of the cells, Ks is the value of the limiting nutrient conc. which results in a growth rate of half the maximum value Ks = 1/2umax. To be explained with a plot of u Vs substrate conc.

3. Illustrate in detail about 'Models of microbial Product formation'

Products formation could be growth associated eg ethanol, Non growth associated eg. Vitamins, antibiotics, partially growth associated eg. Amino acid, Xanthan and lactic acid etc. Luekeking and Piret model to be explained for lactic acid production is a type of partially growth associated type where the production is found to depend on both concentration of cell as well as their growth rate. Hence the expression as $\gamma_p = \alpha\gamma_x + \beta x$; $\gamma_p = \alpha\mu x + \beta x$

Lactic acid is product arising from anaerobic fermentation of sugars such as glucose and lactose. Although the concersion of glucose to lactic acid is direct route for energy production, the production kinetics are not solely growth associated. State the kinetics expression for growth associated, non-growth associated and partially growth associated products formation. For growth associated kinetics the gluconic acid production to be stated and non-growth associated the penicillin production to be stated with expressions.

4. State and explain the kinetic expression for Substrate Inhibited growth phase and Product inhibited growth phase.

Growth inhibited by the substrate can be simply modeled in well stirred tank reactor.

$X+S – XS$ – new cells ; $XS+S$ --- XS_2(non cell complex)

The effect of inhibitory substrate can be considered to result from reversible formation of a complex XS_2 which is no longer to form new cells. We should define the dimensionless variables as per the class notes to find the expression of μ_{max} t value.

Products of cellular metabolism may also inhibit growth and hence slow the rate of their own production. Product inhibition effects on cell growth can be represented in a variety of ways. One common approach is analogous that the substrate inhibition as $\mu = \mu_{max} S/(Ks+S)(1+p/k_p)$ and the second approach is exponential expression $\mu = \mu_{max} S/(Ks+S) e^{-kpP}$. The exponential term is expanded in Taylor series and truncated after first term to achieve the final expression as $\mu = \mu_{max} S/(Ks+S). (1-KpP)$

5. i. Derive and discuss the growth kinetics with plasmid instability

Discuss the problem of plasmid loss – leads to large plasmid free population hence overall reactor efficiency decreases – Plasmid DNA mutation, segregation instability are the reasons – The total number of plasmids in generation must double in the next. A simple model developed for fraction of plasmid bearing cells as the function of time to be derived. The parameters like Probability of plasmid loss per generation of cells, and difference in growth rate of P+ and P- cells populations are used. Finally the Fraction of plasmid bearing cells are $F=1-\alpha-p/1-\alpha-2^{n(+p-1)} p$

(ii) A plasmid containing strain of E.coli to produce recombinant protein in 250 lit fermentors. The probability of plasmid loss in per generation is 0.005. The specific growth rate of P- cells 1.4/h; the specific growth rate of P+ cell is 1.2/h. Estimate the fraction of plasmid bearing cells after 18 hrs of growth. (April/May 2009)

Find n-value and alpha value and substitute in F expression. As per the class notes solving method and the answer should be n=31 generations α value = 1.17 and F value should be 0.45, means that after 18hrs of cultivation 45% of cells will contain the plasmid.

6. Explain Cell mass growth rate & Cell number growth rate. (ii) State and explain the kinetic expression for Substrate Inhibited growth phase. (April/May 2012)

A detailed note on the following i.e Occasionally the double time of Cell number and Cell dry weight may differ hence write as Cell mass growth rate or Specific growth rate (μ hr-1) = $\mu=1/x\ dx/dt$ and Cell number growth rate or Specific cell number growth rate (v hr-1) = $V=1/N\ dn/dt$, Where 'μ ' and 'V' are equal, growth refered to be 'balanced' where there is adequate supply of nutrients, hence the cell composition is constant even though the concentration of all nutrients decreasing. Otherwise in 'unbalance growth' the variation in cell composition, may occur eventhough cell number growth rate is contant, the cell mass growth rate will vary.

7. Explain the Models of Product Inhibition kinetics & replication model. (Nov / Dec 2011)

Products of cellular metabolism may also inhibit growth and hence slow the rate of their own production. Product inhibition effects on cell growth can be represented in a variety of ways. One common approach is analogous that the substrate inhibition as $\mu = \mu_{max} S/(Ks+S)(1+p/k_p)$ and the second approach is exponential expression $\mu = \mu_{max} S/(Ks+S) e^{-kpP}$. The exponential term is expanded in Taylor series and truncated after first term to achieve the final expression as

$\mu = \mu_{max} S/(Ks+S). (1-KpP)$

Plasmid replication model: describes the replication of plasmids within the cells. The number of plasmids within cell, may vary depending on the nature of plasmids and growth rate of host. – The doubling of plasmid number within the cell is governed by two separate factors. The model is as follows. $1/ (\mu_{pwo} = \mu) = K_p/V^{max} (1+ K_p/P_s)$; low mol.wt. plasmids presents in high copy number

8. Write and derive MM equation with its representation of kinetic date types.

As per the class notes the MM equation to be derived with suitable assumptions and considerations such as the 1. at low substrate conc. The reaction velocity is proportional to the substrate conc. and reaction is first order with respect to substrate. 2. As the substrate concentration is increased the reaction rate slows and no longer proportional to substrate concentration, which is mixed order 3. at still high substrate conc. The reaction rate becomes constant and independent of substrate conc. The reaction becomes Zero order. The final MM expression of V= Vmax [S]/{[S]+Km} to be arrived. Then modification of MM equation as Line weaver-Burk plot to be explained with its importance.

9. State and explain the significance of Plasmid Instability with its Growth kinetics model

Plasmid loss through generation, leads to plasmid free population, Plasmid DNA mutation, segregational instability, structural instability; Explain the Growth kinetics with Plasmid Instability and its expression from the class notes - Discuss the problem of plasmid loss – leads to large plasmid free population hence overall reactor efficiency decreases – Plasmid DNA mutation, segregation instability are the reasons – The total number of plasmids in generation must double in the next. A simple model developed for fraction of plasmid bearing cells as the function of time to be derived. The parameters like Probability of plasmid loss per generation of cells, and difference in growth rate of P+ and P- cells populations are used as per the class notes. Finally the Fraction of plasmid bearing cells are $F=1-\alpha-p/1-\alpha-2^{n(+p-1)}p$

10. Write a detailed note on Structured growth model with reference to (i) Compartmental model (ii) Model of Ramakrishna et al.

Structured growth model recognize the complex metabolic reactions occurring within the cell. It also predicts the dynamic behavior of the cell with reference to its environment. The complex organization with in the cell can be described in Genetic system, Epigenetic system, and metabolic system. The models which incorporate the details of intracellular metabolism are referred as "Structured model". These models accounts for unbalanced growth, the composition of the major cellular constituents such as RNA, enzyme conc. etc vary as the result of changing external conditions. State and explain the models like Compartmental model, Ramakrishna et al

11. Illustrate in detail of 'Models of microbial Product formation' (April / May 2011) (Nov 2013)

Products formation could be growth associated eg ethanol, Non growth associated eg. Vitamins, antibiotics, partially growth associated eg. Amino acid, Xanthan and lactic acid etc. Luekeking and Piret model to be explained for lactic acid production by lactobacillus sp. Lactic acid is product arising from anaerobic fermentation of sugars such as glucose and lactose. Although the concersion of glucose to lactic acid is direct route for energy production, the production kinetics are not solely growth associated. State the kinetics expression for growth associated, non-growth associated and partially growth associated products formation. For growth associated kinetics the gluconic acid production to be stated and non-growth associated the penicillin production to be stated with expressions.

12. Write a detailed note on Structured growth model with reference to (i) Model of Cellular energetics (ii) Single cell Model. (November / December 2011)

Structured growth model recognize the complex metabolic reactions occurring within the cell. It also predicts the dynamic behavior of the cell with reference to its environment. The complex organization with in the cell can be described in Genetic system, Epigenetic system, metabolic system. The models which incorporate the details of intracellular metabolism are referred as "Structrured model". These models accounts for unbalanced growth, the compsition of the major cellular constituents such as RNA, enzyme conc. Etc vary as the result of changing external conditions. State and explain the models like, Cellular energitics, Single cell model etc

13. Derive a detailed expression on Microbial growth kinetics under batch cultivation process. (November / December 2011)

Mathematical point of view, there is little difference between enzyme kinetics and cells, (cell metabolism depends on the integrated action of a multitudes of enzymes) Draw the batch growth curve for the microbial growth and mark the difference phases as lag, acceleration phase, log or exponential phase and deceleration phase and stationary phase and death phase. Explain the enzymatic actions during the phases. $\Gamma_x = \mu x$; the fate of μ at the each phase to be noted with reference to μ_{max}. Derive expression as $\ln Xt = \ln Xo + \mu t$ and draw the graph on it. Explain the doubling time also as $td = \ln 2/\mu$.

14. Explain the growth of filamentous organisms and its growth kinetics. (April/May 2012)

15. Find the maximum specific growth (Um) and Monod constant (Ks), following the literalized Monod equation procedure and using the data given in the table for the growth of yeast, on glucose in a 20 L fermenter. The cell growth is reasonably represented by Monod growth kinetics.

Substrate g/L	**:**	**10**	**5**	**3.3**	**2.25**	**2.0**	**1.7**	**1.4**
U (h-1)	**:**	**0.4**	**0.36**	**0.3**	**0.27**	**0.25**	**0.22**	**0.2**

16. Explain the models for substrate and product inhibition. (Nov / Dec 2012) (Nov/Dec 2013)

17. Explain the simple unstructured kinetic models for microbial growth. (Nov / Dec 2012 & 2013)